

Bayesian Workflow in Practice

Simranjeet Bilkhu

May 4, 2026

McMaster University

Background

What is a Statistical Model?

- A statistical model describes how observed data $\mathbf{y}$ is generated, governed by unknown parameters θ
- Inference consists of finding values of θ that best describe the data
- Example: a psychological study where subjects $i = 1, \dots, n$ respond to a questionnaire
- Each subject's responses are represented in a list $\mathbf{y}_i$
- The likelihood $p(\mathbf{y}_1, \dots, \mathbf{y}_n \mid \theta)$ is a family of distributions indexed by θ

Example:

$$p(y_1 \mid \theta) \sim \text{Normal}(\theta, 1)$$

Maximum Likelihood Inference

- If we treat θ as fixed and unknown, we can perform maximum likelihood inference
- The goal is to find θ^{MLE} that maximizes the likelihood of the observed data
- Treats parameters as fixed but unknown constants

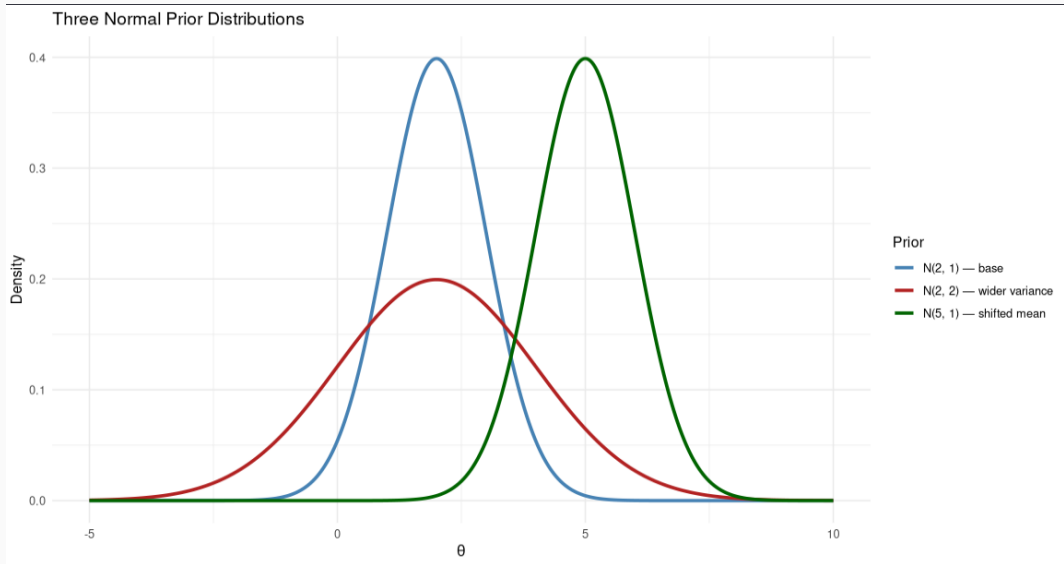
The Bayesian Paradigm

- Since θ is unknown, we can Bayesians would argue to treat it as random
- We impose a prior distribution $p(\theta)$ encoding our beliefs about θ before observing data
- Higher prior probability on a range of θ reflects stronger belief the true value lies there
- Inference is performed via the posterior distribution $\pi(\theta \mid \mathbf{Y})$, derived by Bayes' theorem:

$$\pi(\theta \mid \mathbf{Y}) = \frac{p(\mathbf{Y} \mid \theta) p(\theta)}{\int p(\mathbf{Y} \mid \theta) dp(\theta)}$$

- The denominator is the marginal distribution of $\mathbf{Y}$, often computationally intractable
- Computational algorithms such as MCMC allow us to sample from the posterior up to a proportionality constant

Prior Example



Prior Example Continued

It's usually easy to figure out what the mean parameter is.

Using interval probabilities to interpret priors is a good idea. For instance:

Roughly 95% of the values in a normal distribution lie within two standard deviations of the mean.

Data: King County Home Sales

kingco_sales from the kingCoData R package.

- **605,000+** single family and townhome sales
- **King County, Washington**, 1999 – December 2025
- Some of the variables included:
 - *Transaction*: sale date, assessor warnings, condition
 - *Structure of the House*: sqft, beds, baths, grade, stories, year built, renovations
 - *Location*: latitude/longitude, city, area, zoning, waterfront, views, golf, greenbelt
 - *Tax Assessments*: land and improvement tax values

Outcome of interest: `sale_price`

Data: Modelling Split and Preprocessing

For the purposes of this presentation, I used **Pre-2016** as my *mock* historical data, which I will use to figure out my prior. Then, we will fit a Bayesian Model on the post 2016 data.

Step 1 – Choosing an Initial Model

How do you chose an initial model?

How do you chose an initial model?

- Consulting an Expert.
- Data visualization, and knowledge of the dataset that you are actually working with.
- Previous analysis.
- Need to be careful of data leakage, and data snooping.

Example of Expert Consultation

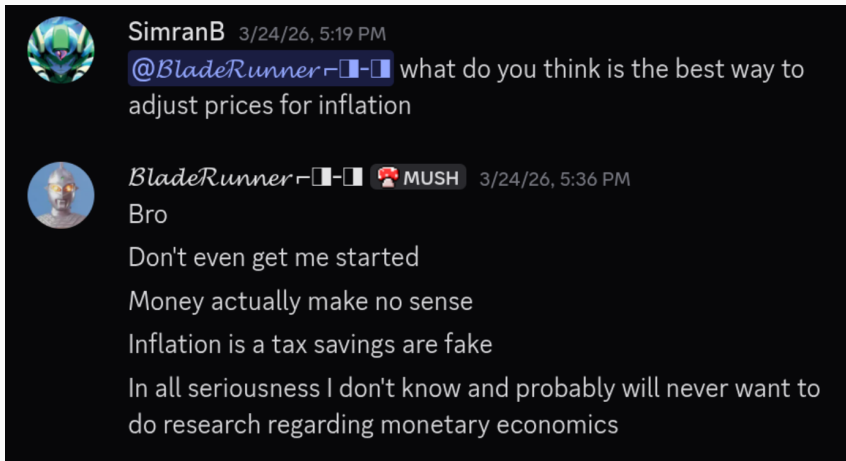


Figure 1: Expert consultation in Practice. BladeRunner is a Master's student in economics.

Step 2 – Prior Specification

It's Important!

- The main criticism of Bayesian statistics is that it isn't purely objective. The prior choice has an effect on the outcome.
- Several approaches have been considered to mitigate this
- Objective Bayes, non-informative Priors.

Objectivity is a myth in any framework. Choose a good prior by using historical/*similar* data and/consult an expert.

Example of Expert Consultation 2



Figure 2: Anna is also a developer for the `lme4` package.

Step 3 – Prior Predictive Check

- Once you have chosen your prior, you should validate it.
- The approach is to simulate data from $p(y)$ (the marginal likelihood of your data) and see if it makes sense.
- To understand where the prior comes in, note drawing $\theta \sim p(\theta)$ and then $y \sim p(y \mid \theta)$ is the same as drawing $(y, \theta) \sim p(y, \theta)$.

Step 4 – Fitting the Model

Fitting the Model

There are many computational algorithms to fit bayesian models. Usually there is a tradeoff between speed and accuracy:

- Laplace Approximation.
- Variational Inference.
- MCMC

We will be using tools built on Stan: <https://mc-stan.org/>.

WARNING: Stan programs can take a WHILE to compile, and are pretty resource intensive (can take up lots of RAM while compiling). Plan accordingly.

Goal: Sample from a complex posterior distribution $p(\theta|y)$

- Construct a Markov chain that has $p(\theta|y)$ as its stationary distribution
- Simulated chain for a long time
- Use the chain's samples as draws from $p(\theta|y)$

Variational Inference

- Variational inference (VI) is an alternative to MCMC for approximating the posterior $p(\theta | y)$
- Instead of sampling, VI turns inference into an optimization problem
- We choose a family of distributions $q(\theta)$ and find the member closest to the true posterior
- Closeness is measured by KL divergence, minimized by maximizing the ELBO
- Faster than MCMC but approximate

Step 5 – Validating Computation

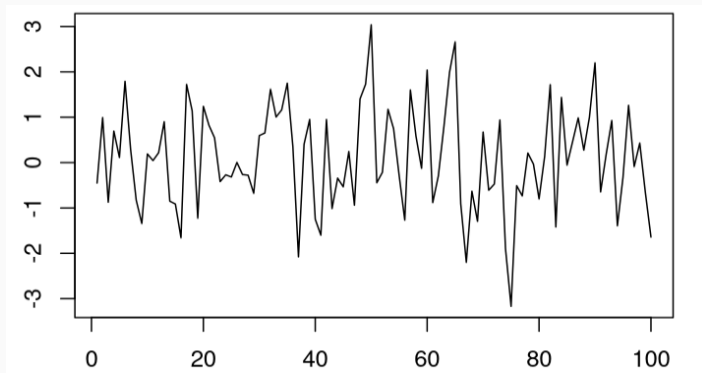


Figure 3: Example of a “good” trace plot

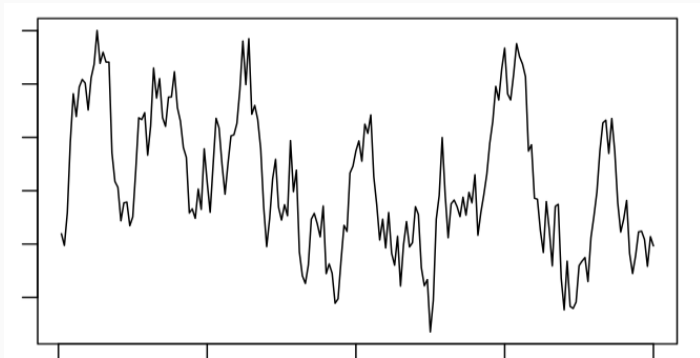


Figure 4: Example of a “bad” trace plot

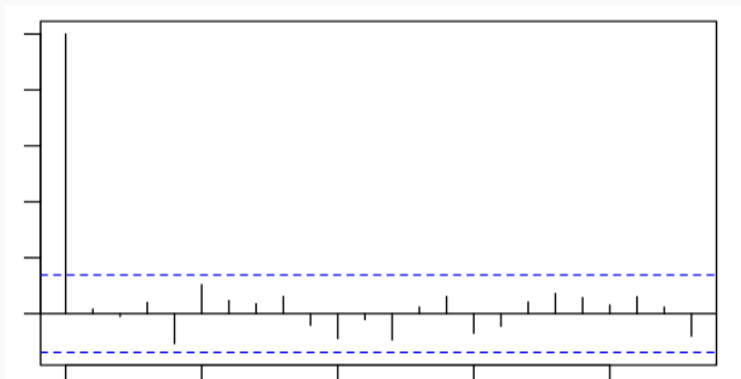


Figure 5: Example of a “good” acf plot

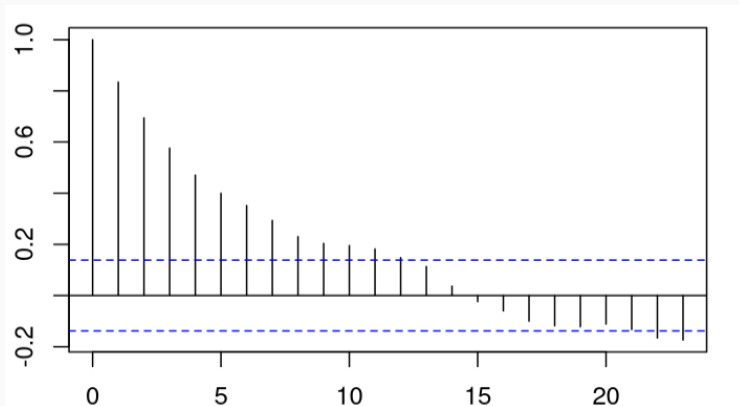


Figure 6: Example of a “bad” acf plot

There are other numerical diagnostics that can flag a problem.

- $\hat{R}$
- ESS
- Geweke's
- ...

Look at: <https://proceedings.mlr.press/v80/yao18a/yao18a.pdf>

Step 6 – Posterior Predictive Check

Posterior Predictive Check

- We simulate replicated data $y_{rep} \sim p(y_{rep} | y)$ from the posterior predictive distribution.
- To do so, we first draw $\theta \sim \pi(\theta | y)$ from the posterior, then draw $y_{rep} \sim p(y | \theta)$ from the likelihood.
- The idea is to compare how this simulated data looks compared to the observed data.
- Gives us a good idea of how suitable our model is for this data.

Step 7 – Modifying the Model

Problems with the posterior predictive check, or simulation usually mean that you have to adjust the model.

- If you have a problem with posterior predictive check, try changing your priors (such as using a weakly informative prior), and adjust your model.
- Same thing applies. You can also try using different algorithms as well.

More on Model Revision

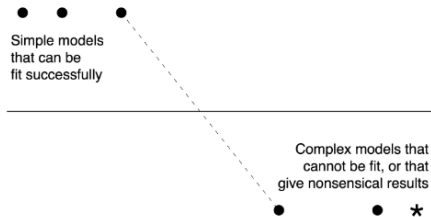


Figure 11: *Diagram of advice for debugging. The asterisk on the lower right represents the scenario in which problems arise when trying to fit the desired complex model. The dots on the upper left represent successes at fitting various simple versions, and the dots on the lower right represent failures at fitting various simplifications of the full model. The dotted line represents the idea that the problems can be identified somewhere between the simple models that fit and the complex models that don't. From Gelman and Hill (2007).*

Figure 7: A useful diagram for debugging computational problems.

Final Remarks

This was a very brief overview of a Bayesian workflow.

In general, you should try to consider many different candidate models, and compare.

- Bayesian Model averaging
- Sensitivity analysis
- Comparing different computational methods (when possible)

The most important thing is transparency. Keep track of the decisions you are making.

A really good reference is the Stan User Guide:

<https://mc-stan.org/docs/stan-users-guide/>

Take a look at this paper for a lengthy overview of a bayesian workflow:

<https://arxiv.org/pdf/2011.01808>

King County Data: <https://github.com/andykrause/kingCoData>